Biquadratic equations and equations reducible to quadratics

One substitution turns a degree-four equation, or a fractional one, into a quadratic you can already solve.

LESSON 70–71 | 2 × 40 MIN | ALGEBRA 8, CHAPTER III §24 | STAGE 9 · BEYOND

LESSON CLOCK

5' | 12' | 8' | 13' | 2'

Warm-up	5 min
Explanation	12 min
Interactive	8 min
Practice	13 min
Homework	2 min

LEARNING OBJECTIVES

- Solve a biquadratic equation by the substitution $t = x^2$.
- Reject negative values of t and explain why.
- Solve a fractional-rational equation that reduces to a quadratic.
- Check for values excluded by the denominator.

TEXTBOOK

National	Chapter III, pp. 147–152
Cambridge	Extension beyond Stage 9
Workbook	–

01 Explanation

Biquadratic equations

DEFINITION

An equation of the form $ax^4 + bx^2 + c = 0$ is called **biquadratic**. Only even powers appear.

Put $t = x^2$. Since $x^4 = (x^2)^2 = t^2$, the equation becomes

$$at^2 + bt + c = 0$$

— an ordinary quadratic. Solve it, then **return to x** by solving $x^2 = t$ for each value of t found.

THE STEP EVERYONE FORGETS

$t = x^2$ is a square, so $t \geq 0$. Any **negative** value of t must be **rejected** — it gives no real x. And every positive t gives **two** values of x, namely $\pm\sqrt{t}$.

So a biquadratic equation can have 4, 3, 2, 1 or 0 real roots — count carefully.

A worked pattern

$$x^4 - 13x^2 + 36 = 0$$

$$t^2 - 13t + 36 = 0 \ \ \text{(with } t = x^2\text{)}$$

$$t = 4 \text{ or } t = 9 \ \ \text{(sum 13, product 36)}$$

$$x^2 = 4 \Longrightarrow x = \pm 2 \quad x^2 = 9 \Longrightarrow x = \pm 3$$

Four roots: $-3, -2, 2, 3$.

Fractional equations that reduce to quadratics

Multiply through by the common denominator, then solve the quadratic that appears — but list the excluded values **first**.

$$\frac{x}{x-2} + \frac{4}{x} = 3, \quad x \neq 0, x \neq 2$$

Multiplying by $x(x - 2)$:

$$x^2 + 4(x - 2) = 3x(x - 2)$$

$$x^2 + 4x - 8 = 3x^2 - 6x \Longrightarrow 2x^2 - 10x + 8 = 0 \Longrightarrow x^2 - 5x + 4 = 0$$

$$x = 1 \text{ or } x = 4 \text{ — both permissible ✓}$$

CHECK AGAINST THE EXCLUDED LIST

If a root turns out to be an excluded value it is an **extraneous root** and must be thrown away. Multiplying by an expression that could be zero is exactly what creates them.

02 Worked examples

EXAMPLE 1 ***Solve $x^4 - 5x^2 + 4 = 0$***

1. $t = x^2$: $t^2 - 5t + 4 = 0$

   The substitution.

2. $t = 1$ or $t = 4$

   Sum 5, product 4.

3. $x^2 = 1 \Longrightarrow x = \pm 1$

4. $x^2 = 4 \Longrightarrow x = \pm 2$

   Four roots in all.

ANSWER

$x = \pm 1,\ x = \pm 2$

EXAMPLE 2 ***Solve $x^4 + 3x^2 - 4 = 0$***

1. $t^2 + 3t - 4 = 0$

2. $t = 1$ or $t = -4$

3. $t = -4$ is rejected — a square cannot be negative.

   This is the key step.

4. $x^2 = 1 \Longrightarrow x = \pm 1$

   Only two roots.

ANSWER

$x = \pm 1$

EXAMPLE 3 **Solve $\frac{6}{x} = x - 1$**

1. $x \neq 0$

   Note the exclusion first.

2. $6 = x(x - 1)$

   Multiply by x.

3. $x^2 - x - 6 = 0$

4. $x = 3$ or $x = -2$

   Neither is 0, so both stand.

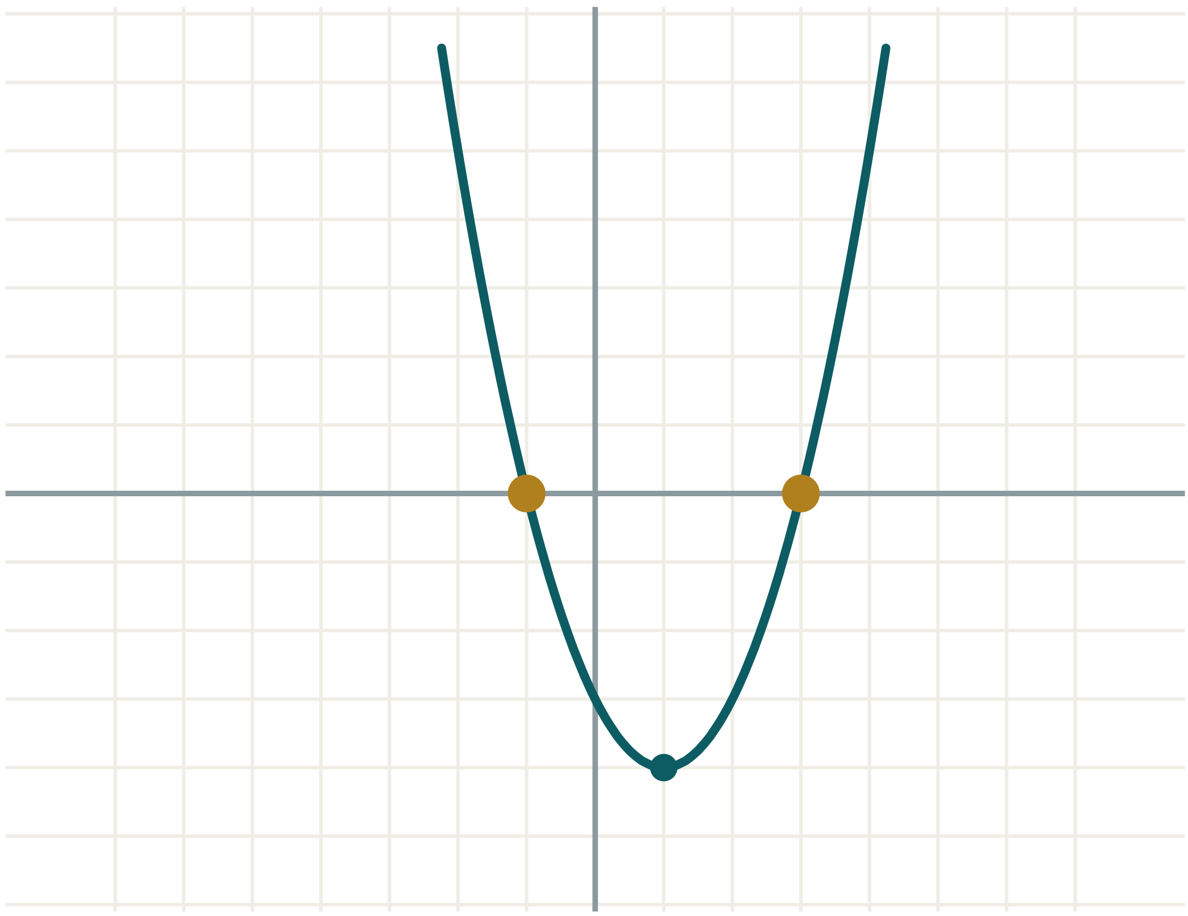
ANSWER

$x = 3, x = -2$

03 Interactive model

Use the model on the quadratic in t, then ask the class how many x-values each t produces.

The quadratic in t



a **1** | b **-2** | c **-3** | $D = b^2 - 4ac$ **16** | roots **two** | x_1 **-1** | x_2 **3** | $x_1 + x_2$ **2** | $x_1 \cdot x_2$ **-3**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Biquadratic equation	Bikvadrat tenglama	Биквадратное уравнение
Substitution	Almashtirish	Замена переменной
New variable	Yangi o‘zgaruvchi	Новая переменная
Reject a value	Qiymatni rad etish	Отбросить значение
Fractional-rational equation	Kasr-ratsional tenglama	Дробно-рациональное уравнение
Extraneous root	Chet ildiz	Посторонний корень
Return to x	x ga qaytish	Вернуться к x
Degree of an equation	Tenglamaning darajasi	Степень уравнения

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

For a biquadratic the substitution is:

- $t = x$
- $t = x^2$
- $t = x^4$
- $t = \sqrt{x}$

If $t = -9$ comes out of the substitution, you should:

- $take\ x = \pm 3$
- reject it
- $take\ x = -3$
- start again

$x^4 - 13x^2 + 36 = 0$ has how many real roots?

- 2
- 3
- 4
- 0

Before multiplying a fractional equation by its denominator you must:

- square both sides
- list the excluded values
- factorise the numerator
- nothing

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Solve $x^4 - 5x^2 + 4 = 0$

ANSWER $x = \pm 1, \pm 2$

2. Solve $x^4 - 10x^2 + 9 = 0$

ANSWER $x = \pm 1, \pm 3$

3. Solve $x^4 - 16 = 0$

ANSWER $x = \pm 2$

4. Solve $x^4 - 9x^2 = 0$

ANSWER $x = 0, \pm 3$

5. If $t = x^2$ and $t = 25$, find x.

ANSWER $x = \pm 5$

6. If $t = x^2$ and $t = -1$, find x.

ANSWER No real x.

7. Solve $\frac{4}{x} = 2$

ANSWER $x = 2$

Medium · the standard question

7 PROBLEMS

1. Solve $x^4 + 3x^2 - 4 = 0$

ANSWER $x = \pm 1$

2. Solve $x^4 - 13x^2 + 36 = 0$

ANSWER $x = \pm 2, \pm 3$

3. Solve $x^4 + 5x^2 + 4 = 0$

ANSWER No real roots — both t values are negative.

4. Solve $\frac{6}{x} = x - 1$

ANSWER $x = 3, x = -2$

5. Solve $\frac{12}{x} + x = 7$

ANSWER $x = 3, x = 4$

6. Solve $x^4 - 8x^2 + 16 = 0$

ANSWER $t = 4$ twice, so $x = \pm 2$

7. Solve $\frac{1}{x} + \frac{1}{x+1} = \frac{5}{6}$

ANSWER $x = 2$ or $x = -3$

Hard · stretch and reason

7 PROBLEMS

1. *Solve* $4x^4 - 17x^2 + 4 = 0$

   ANSWER $t = 4$ or $t = 0.25$: $x = \pm 2, \pm 0.5$

2. *Solve* $\dfrac{x}{x-2} + \dfrac{4}{x} = 3$

   ANSWER $x = 1, x = 4$

3. *Solve* $\dfrac{x+1}{x-1} - \dfrac{x-1}{x+1} = 1$

   ANSWER $4x = x^2 - 1$: $x = 2 \pm \sqrt{5}$

4. *Solve* $(x^2 - 1)^2 - 5(x^2 - 1) + 4 = 0$

   ANSWER Let $t = x^2 - 1$: $t = 1$ or 4, so $x = \pm\sqrt{2}, \pm\sqrt{5}$

5. *For which k does* $x^4 + kx^2 + 4 = 0$ *have four real roots?*

   ANSWER Need two positive t-roots: $k^2 > 16$ and $-k > 0$, so $k < -4$.

6. *Solve* $\dfrac{2}{x-3} = \dfrac{x}{x-3} - 1$

   ANSWER $x \neq 3$; $2 = x - (x - 3) = 3$ — false, so no solutions.

7. *Solve* $\dfrac{x^2 - 4}{x - 2} = 5$

   ANSWER $x \neq 2$; $x + 2 = 5$, so $x = 3$.

07 Homework

Homework — 6 problems

Algebra 8, Chapter III, pp. 147–152. Reject negative t values and say so in writing.

1. *Solve* $x^4 - 26x^2 + 25 = 0$
2. *Solve* $x^4 + 2x^2 - 3 = 0$
3. *Solve* $x^4 + 4x^2 + 3 = 0$

4. *Solve* $\frac{8}{x} = x + 2$

5. *Solve* $\frac{20}{x} - x = 1$

6. *Solve* $x^4 - 4x^2 = 0$